

What is Keras?

Keras is an open-source, high-level Deep Learning API used to build and train Artificial Intelligence (AI) and Deep Learning (DL) models. It is integrated with TensorFlow and provides a simple interface for creating neural networks.

Developed by: François Chollet

Released: 2015

Backend: TensorFlow (default)

Website: <https://keras.io>

Features of Keras

- Open-source and free
- Easy to learn and use
- High-level API for Deep Learning -Built on top of TensorFlow
- Supports CPU, GPU, and TPU
- Fast model development
- Supports Convolutional Neural Networks (CNN)
- Supports Recurrent Neural Networks (RNN)
- Supports Transfer Learning
- Large community support

Advantages of Keras

- Beginner-friendly
- Less coding required
- Easy model building
- Fast prototyping
- Excellent documentation
- Supports deep learning - applications
- Integrates with TensorFlow

Disadvantages of Keras

- Less flexible than low-level TensorFlow
- Limited control over complex operations
- Depends mainly on TensorFlow backend

Installation

```
pip install keras
```

Import Keras

```
import keras

Using TensorFlow Keras

from tensorflow import keras

Check Version

import keras

print(keras.__version__)
```

Creating a Sequential Model

```
from tensorflow import keras

model = keras.Sequential()
```

Adding Layers

```
from tensorflow.keras.layers import Dense

model.add(Dense(64, activation="relu")) model.add(Dense(10, activation="softmax"))
```

Common Layers

Dense Conv2D MaxPooling2D Flatten Dropout LSTM Embedding BatchNormalization

Activation Functions

ReLU Sigmoid Softmax Tanh Linear

Example

```
Dense(64, activation="relu")
```

Compile Model

```
model.compile( optimizer="adam", loss="categorical_crossentropy", metrics=["acc
```

Train Model

```
model.fit( X_train, y_train, epochs=10, batch_size=32 )
```

Evaluate Model

```
model.evaluate( X_test, y_test )
```

Prediction

```
model.predict(X_test)
```

Save Model

```
model.save("model.keras")
```

Load Model

```
from tensorflow.keras.models import load_model model = load_model("model.keras")
```

Optimizers

Adam SGD RMSprop Adagrad Adamax

Example

```
optimizer="adam"
```

Loss Functions

- Mean Squared Error (MSE)
- Binary Crossentropy
- Categorical Crossentropy Sparse Categorical - - - Crossentropy

Metrics

- Accuracy
- Precision
- Recall
- Mean Absolute Error (MAE)

Callbacks

Common Callbacks

- EarlyStopping
- ModelCheckpoint-
- ReduceLROnPlateau

Example

```
from tensorflow.keras.callbacks import EarlyStopping callback = EarlyStopping()
```

✓ Common Keras Functions

Function	Purpose
<code>keras.Sequential()</code>	Create a sequential model
<code>model.add()</code>	Add layers to the model
<code>Dense()</code>	Create a fully connected (Dense) layer
<code>model.compile()</code>	Configure the model before training
<code>model.fit()</code>	Train the model using training data
<code>model.evaluate()</code>	Evaluate the model performance on test data
<code>model.predict()</code>	Predict the output for new input data

Function	Purpose
<code>model.save()</code>	Save the trained model to a file
<code>load_model()</code>	Load a previously saved model
<code>keras.Input()</code>	Define the input layer of the model

Applications of Keras

- Deep Learning
- Artificial Intelligence
- Image Classification
- Object Detection
- Face Recognition
- Speech Recognition
- Natural Language - - Processing (NLP)
- Chatbots
- Recommendation Systems
- Medical Diagnosis
- Financial Forecasting
- Robotics

✓ Keras vs TensorFlow

Keras	TensorFlow
High-level API	Complete Deep Learning Framework
Easy to learn and use	More advanced and feature-rich
Requires less coding	Requires more coding
Fast model development	More flexible and customizable
Best for beginners	Best for complex AI applications
Runs on TensorFlow	Includes Keras as its official high-level API
Focuses on simplicity	Provides complete control over deep learning models
Ideal for rapid prototyping	Ideal for large-scale production systems
Easier to build neural networks	Supports advanced model customization
Suitable for small and medium deep learning projects	Suitable for research and enterprise-level AI applications

Future of Keras

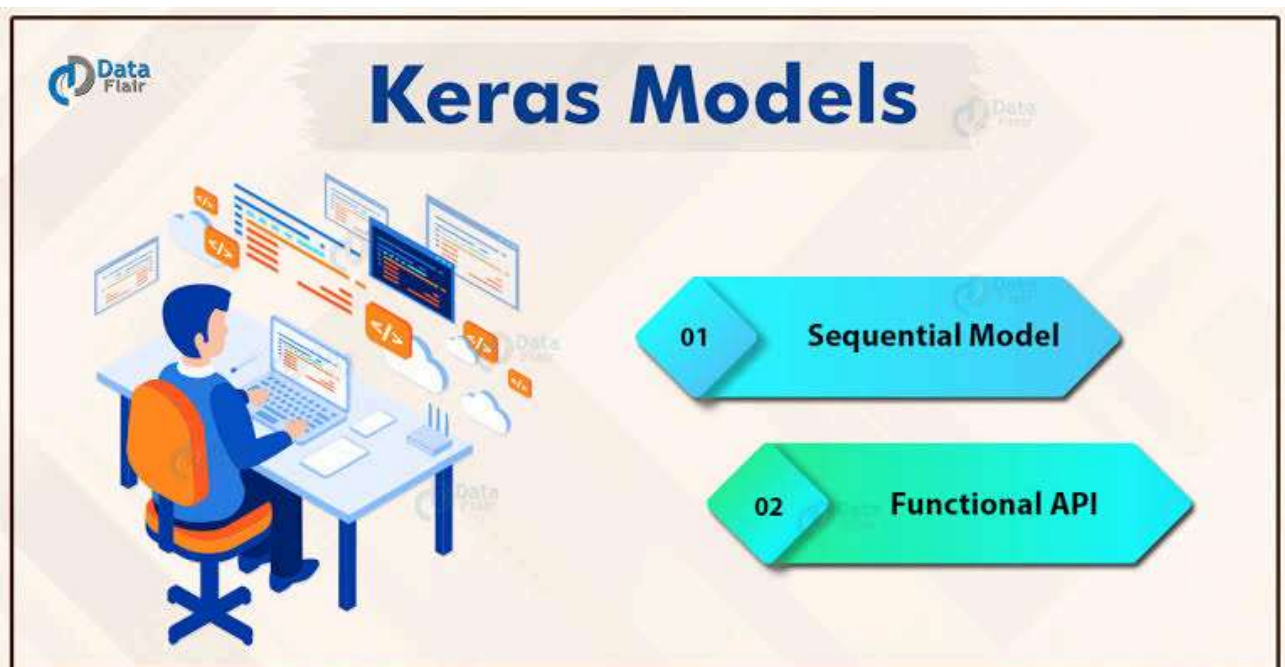
1. Greater adoption in Artificial Intelligence and Deep Learning.
2. Better integration with TensorFlow.
3. Faster model development tools.

4. Improved support for cloud and mobile deployment.
5. Better optimization for GPUs and TPUs.
6. Growth in healthcare, finance, and robotics.
7. Easier development of advanced neural networks.
8. Continuous updates from the open-source community.
9. Increased use in education and research.
10. Growing demand in machine learning and AI industries.

Conclusion

Keras is a powerful, user-friendly, and high-level Deep Learning API that simplifies the process of building, training, and deploying neural network models. Built on top of TensorFlow, it provides an easy interface for creating AI applications with minimal code. Keras is widely used in image processing, natural language processing, healthcare, finance, robotics, and many other Artificial Intelligence applications, making it one of the most popular deep learning libraries for beginners and professionals.

keras Models



Keras Future

